

Shiv Nadar University Chennai

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

Experiment 1 Implementation of a Single Layer Perceptron for Binary Classification

1. Objective

The objective of this experiment is to understand the concept of an artificial neuron and implement a Single Layer Perceptron from scratch for binary classification. Students will learn the perceptron learning algorithm, understand the importance of activation functions, visualize the learning process, and evaluate the classifier using a real-world dataset.

2. Tasks to be Performed

1. Download and understand the dataset.
2. Perform Exploratory Data Analysis (EDA).
3. Preprocess the dataset.
4. Implement a Single Layer Perceptron from scratch.
5. Train the model using the perceptron learning rule.
6. Evaluate the classifier using standard performance metrics.
7. Analyze the effect of different learning rates.
8. Compare the implementation with Scikit-learn's Perceptron (optional).

3. Background Theory

An Artificial Neuron is the fundamental building block of a neural network. It receives one or more input features, multiplies them by corresponding weights, adds a bias term, and applies an activation function to determine the output.

The Single Layer Perceptron is the earliest supervised learning algorithm capable of solving linearly separable binary classification problems.

Mathematical Formulation

Weighted Sum

$$z = \mathbf{w}^T \mathbf{x} + b$$

where

- $\mathbf{x}$: Input vector
- $\mathbf{w}$: Weight vector
- b : Bias
- z : Linear combination

Prediction

$$\hat{y} = f(z)$$

Weight Update Rule

$$\mathbf{w}_{new} = \mathbf{w}_{old} + \eta(y - \hat{y})\mathbf{x}$$

Bias Update

$$b_{new} = b_{old} + \eta(y - \hat{y})$$

where η denotes the learning rate.

4. Activation Functions

An activation function determines the output of a neuron based on the weighted sum of its inputs. It introduces non-linearity, enabling neural networks to learn complex patterns. Although the perceptron uses only the Step Activation Function, modern deep learning models employ differentiable activation functions for efficient optimization through backpropagation.

Step Activation Function

$$f(z) = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

The Step Activation Function produces a binary output and is therefore suitable only for linearly separable binary classification problems.

Activation	Output Range	Typical Applications
Step	$\{0, 1\}$	Classical Perceptron
Sigmoid	$(0, 1)$	Binary Classification Output Layer
Tanh	$(-1, 1)$	Hidden Layers (Traditional Networks)
ReLU	$[0, \infty)$	Hidden Layers in CNNs and MLPs
Leaky ReLU	$(-\infty, \infty)$	Avoiding Dying ReLU
Softmax	$(0, 1)$	Multi-class Classification Output Layer

Note: This experiment uses only the Step Activation Function. The remaining activation functions will be studied in subsequent experiments involving Multilayer Perceptrons and Deep Neural Networks.

5. Dataset Description

Dataset: Banknote Authentication Dataset

Source: UCI Machine Learning Repository

Download Link:

<https://archive.ics.uci.edu/dataset/267/banknote+authentication>

Instances	1372
Features	4 Numerical Features
Classes	2
Missing Values	None
Task	Binary Classification

Feature Description

- Variance
- Skewness
- Curtosis
- Entropy

Target Class

- 0 – Authentic Banknote
- 1 – Forged Banknote

6. Experimental Procedure

Task 1: Dataset Exploration

- Load the dataset.
- Display the first five samples.
- Determine dataset dimensions.
- Identify missing values.
- Display descriptive statistics.

Expected Output

Dataset summary and statistical information.

Task 2: Exploratory Data Analysis

Generate

- Histograms

- Correlation Heatmap
- Scatter Plot
- Boxplots

Task 3: Data Preprocessing

- Normalize all numerical features.
- Split the dataset into Training (80%) and Testing (20%).

Task 4: Perceptron Implementation

Implement from scratch

- Weight Initialization
- Bias Initialization
- Step Activation Function
- Forward Propagation
- Perceptron Learning Rule

Task 5: Model Training

Display

- Epoch Number
- Misclassified Samples
- Updated Weights
- Updated Bias

Task 6: Model Evaluation

Compute

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

7. Mandatory Plots

Plot	Purpose	Expected Inference
Feature Histograms	Distribution Analysis	Identify skewness and outliers.
Correlation Heatmap	Feature Relationship	Observe highly correlated features.
Scatter Plot	Class Distribution	Determine whether classes are approximately linearly separable.
Training Error vs Epoch	Learning Behaviour	Verify convergence of the perceptron.
Weight Evolution	Parameter Analysis	Observe stabilization of learned weights.
Bias Evolution	Parameter Analysis	Understand the role of the bias term.
Confusion Matrix	Performance Evaluation	Interpret TP, FP, TN and FN.
Learning Rate Comparison	Hyperparameter Study	Compare convergence behaviour for different learning rates.
Decision Boundary (Optional)	Visualization	Illustrate the separating hyperplane using two selected features.

Students should provide a brief interpretation (2–3 sentences) for every generated plot.

8. Performance Tables

Training Summary

Dataset Size	4 features x 1732 instances
Train/Test Split	80 - 20 ratio
Learning Rate	0.01 (Doesnt affect misclassified samples as step function is not differentiable)
Epochs	10 (6th Epoch results in least number of misclassified samples)
Final Weights	[-1.0268 -1.2769 -1.1338 -0.0096]
Final Bias	-0.6
Accuracy	0.9964
Precision	0.9919
Recall	1.0000
F1-score	0.9959

Epoch-wise Learning

Epoch	Errors	Weight 1	Weight 2	Bias
1	59	-0.6803	-0.9381	-0.3
2	25	-0.7754	-1.0421	-0.4
3	20	-0.8021	-1.1266	-0.4
4	19	-0.8828	-1.1959	-0.5
5	20	-0.9909	-1.2951	-0.5
6	17	-1.0268	-1.2769	-0.6
7	24	-1.0813	-1.3928	-0.6
8	22	-1.1702	-1.3568	-0.6
9	20	-1.1755	-1.4696	-0.6
10	20	-1.3069	-1.5412	-0.6

9. Additional Tasks

1. Compare the Step and Sigmoid activation functions. Discuss why the Step Activation Function is not used in modern deep learning models.
2. Compare the implementation with Scikit-learn's Perceptron.
3. Repeat the experiment using learning rates 0.001, 0.01 and 0.1.
4. Explain why a Single Layer Perceptron cannot solve the XOR problem.
5. Study the effect of feature normalization on model convergence.

10. Report Contents

The report should include:

- Objective
- Background Theory
- Activation Functions
- Dataset Description
- Experimental Procedure
- Source Code
- Experimental Results
- Generated Plots with Interpretation

- Performance Tables
- Discussion
- Conclusion
- References

11. References

1. F. Rosenblatt, “The Perceptron,” *Psychological Review*, 1958.
2. I. Goodfellow, Y. Bengio and A. Courville, *Deep Learning*, MIT Press, 2016.
3. C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
4. S. Haykin, *Neural Networks and Learning Machines*, Pearson, 2009.
5. UCI Machine Learning Repository – Banknote Authentication Dataset.
6. Scikit-learn Documentation: Perceptron.